

Cybersecure and Cyber-Resilient European Cloud–Edge–IoT Computing Continuum: A Strategic Roadmap

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ABSTRACT The distribution of computation across cloud, edge, and IoT tiers, combined with the simultaneous operationalisation of NIS2, the Cyber Resilience Act, the AI Act, and DORA, has made cybersecurity, resilience, and digital sovereignty an integrated strategic problem for Europe. Perimeter-oriented security models do not scale to heterogeneous, physically exposed, and multi-jurisdictional continuum infrastructures. This paper addresses that problem by proposing a two-stage methodology for identifying and prioritising cybersecurity capabilities: architectural decomposition of the continuum into six Security Layers and four Transversal Security Cross-Layers together with their constituent capabilities, followed by priority clustering and phased transition-matrix analysis through a seven-dimension template analysis. Applying the methodology, the paper identifies ten strategic capabilities spanning hardware roots of trust, post-quantum cryptographic agility, lightweight edge security, federated interoperability, supply chain verification, cognitive security operations, pan-European network federation, confidential computing, resilient infrastructure, and proactive threat management. These are consolidated into three executive priority clusters, Sovereign Trusted Infrastructure, Cognitive Security and Autonomous Cyber Defence, and Federated Operational Sovereignty, analysed over a 2026–2040 transition horizon. The contribution is both methodological, providing a reproducible template for cybersecurity roadmap construction, and strategic, framing autonomous, federated, and certified Security-by-Design as the operational target for European digital infrastructure.

INDEX TERMS Certification, Cloud–Edge–IoT continuum, confidential computing, cyber-resilience, digital sovereignty, European Union, post-quantum cryptography, security roadmap, Zero-Trust architecture.

I. INTRODUCTION

The shift from centralised cloud to a distributed Cloud–Edge–IoT continuum relocates computation, storage, and inference closer to devices, users, and physical processes [1]–[3]. The architectural gains (lower latency, higher operational autonomy, and locality of data processing) come at a structural cost: attack surface, system heterogeneity, and the number of administrative and trust domains that a defender must reason about simultaneously all increase sharply, while the perimeter-oriented and siloed security models designed for consolidated data-centre environments become structurally inadequate for infrastructures that are geographically dispersed, resource-constrained, and physically exposed [4]–[6]. Each relocation of workload across providers and juris-

dictions is also a transfer of control, which makes continuum security inseparable from digital sovereignty, that is, the ability to govern, audit, and operate critical infrastructure under credible European influence. This coupling is sharpened by concurrent global trends, namely hyperscaler concentration in the EU cloud market [7], the migration to post-quantum cryptography [8], and the weaponisation of generative AI against both human operators and AI-enabled defences, and by the European regulatory response, which translates these pressures into binding obligations on essential entities, products with digital elements, financial services, and high-risk AI systems [9]–[12].

Europe faces this transition under four simultaneous regulatory activations. NIS2 [9], the Cyber Resilience Act (CRA)

[10], the AI Act [12], and DORA [11] together raise the baseline of mandatory security, resilience, accountability, and supply chain obligations across essential entities, ICT products, high-risk AI systems, and financial sector third-party providers. The EU cybersecurity certification framework [13], [14] provides the assurance substrate these regulations presuppose. Operationalisation of this combined regime is uneven: national transposition lags the legal timetable in several large Member States [15], EUCS remains unadopted [16], and conformity assessment infrastructure for AI-specific security obligations is not yet in place [17].

The central argument of this paper is that Europe's normative leadership in trust, privacy, governance, and certification [18], [19] translates into strategic advantage only when it is embedded in deployable capabilities. We call the distance between normative leadership and deployable capacity the ambition–implementation gap. Closing this gap requires a transition from reactive, siloed security towards autonomous, federated, and certified Security-by-Design: infrastructures that sustain trustworthy operation under adverse conditions (autonomous), security mechanisms that span providers and jurisdictions (federated), and assurance embedded through the system lifecycle (certified Security-by-Design) [4], [20], [21].

In this paper, a roadmap is understood as a structured strategic instrument that links current capability gaps to future target states through phased actions, dependencies, and transition conditions. It is not a forecast of technological development, but a coordination tool for aligning research, innovation, policy, standardisation, deployment, and ecosystem investment over time [3]. The Cloud–Edge–IoT continuum refers to the distributed execution environment in which computation, storage, networking, sensing, and AI inference are dynamically placed across central cloud platforms, intermediate edge infrastructures, and resource-constrained devices [1]–[3]. The term continuum is used because these tiers are increasingly interdependent rather than isolated: workloads, data flows, trust decisions, and control functions move across them according to latency, resilience, privacy, energy, and sovereignty requirements [1], [22]. The term ecosystem captures the fact that secure operation depends not only on technologies, but also on providers, standards bodies, regulators, certification schemes, open-source communities, research infrastructures, SMEs, large industrial actors, and public authorities whose decisions jointly determine deployment capacity [3], [13], [14].

The paper makes four contributions.

- First, a two-stage methodology for constructing cybersecurity roadmaps under conditions of ecosystem uncertainty, combining architectural decomposition (Stage 1) with priority clustering and phased transition-matrix analysis (Stage 2), and integrating a Sovereignty Level construct as an interpretive strategic lens (Section III).
- Second, an architectural decomposition of the Cloud–Edge–IoT continuum into six Security Layers and four Transversal Security Cross-Layers, populated with their

constituent capabilities, as the Stage 1 output (Section IV).

- Third, a consolidation of those capabilities into ten strategic priorities grouped in three priority clusters (Sovereign Trusted Infrastructure; Cognitive Security and Autonomous Cyber Defence; Federated Operational Sovereignty), each analysed through a seven-dimension template and projected onto a three-phase transition matrix over 2026–2040 (Section V).
- Fourth, a set of implementation-oriented recommendations derived from the cluster analysis and anchored in the current EU regulatory and industrial landscape, identifying concrete Phase-1 critical enablers (e.g. EUCC “High” certification of an EU-origin RISC-V secure element; OASC MIM6 progression; a standardised adversarial robustness test suite for security AI) whose absence or delivery determines whether the 2040 target state is reachable (Section VI).

Section II provides the strategic and regulatory context. Section III presents the two-stage methodology. Section IV defines the layered system model and its constituent capabilities. Section V presents the three priority clusters and their transition matrices. Section VI states the recommendations. Section VII discusses validation and limitations. Section VIII concludes.

II. BACKGROUND AND DRIVING FACTORS

The first vector shaping the continuum transition is architectural: the continuum is characterised by decentralisation and physical exposure. Decentralisation complicates visibility, orchestration, patching, identity, and incident response across multiple administrative domains and increases the probability of inconsistent security postures across deployments [23], [24]. Physical exposure removes the environmental and access controls that hyperscale data centres rely on: edge nodes in factories, substations, transport hubs, and telecom sites are exposed to tampering, fault injection, side-channel attacks, and supply chain compromise [6]. Hardware roots of trust, secure boot, remote attestation, and tamper-aware design are therefore baseline requirements rather than optional enhancements [25], [26].

The European regulatory landscape constitutes a second major vector shaping the continuum transition. NIS2 [9], the CRA [10], the AI Act [12], DORA [11], the Data Act [27], and GDPR [28] together impose obligations that must be translated into architectures, assurance mechanisms, and lifecycle controls operating across cloud, edge, and IoT. This simultaneity creates disproportionate implementation pressures for SMEs and smaller Member States.

Sovereignty is the third vector. Dependence on non-EU providers for secure silicon, cloud platforms, and enterprise security tooling creates geopolitical and operational risk. Extra-territorial data access under foreign jurisdiction has been confirmed as a practical constraint for EU users of hyperscaler services [29].

Europe's position in this landscape is asymmetric. A world-leading regulatory foundation and a rigorous certification framework [13], [14] coexist with structural industrial gaps. EU-headquartered vendors (Infineon, NXP, STMicroelectronics) design the dominant secure element products and achieved EUCC "High" certification for multiple product families in 2025 [30], but fabrication depends on non-EU foundries and designs remain proprietary, preventing sovereign auditability at the silicon level. Intel SGX/TDX, AMD SEV-SNP, and Arm TrustZone/CCA constitute the dominant operational hardware-backed TEE families across cloud/server and embedded deployments; EU-origin RISC-V alternatives remain largely at research or prototype stage [31]. [No EU-origin RISC-V processor has received EUCC or Common Criteria certification as of April 2026. The enterprise SIEM market is dominated by US platforms, with no EU-headquartered vendor in the Gartner Leaders quadrant as of late 2025 \[32\]. Hyperscalers hold approximately 70% of the EU cloud market; EU-headquartered providers collectively hold about 15%, stable since 2022 but below the 29% share recorded in 2017 \[7\]. For post-quantum migration the NIS Cooperation Group roadmap \[8\] now provides a binding framework, but production-scale post-quantum cryptographic \(PQC\) hardware tailored for edge and IoT is not yet available in Europe.](#)

III. METHODOLOGY

The methodology examines the heterogeneous landscape of the Cloud–Edge–IoT ecosystem and translates it into a structured roadmap for research, innovation, policy, deployment, and ecosystem development. It proceeds in two stages, followed by iterative review and refinement. Stage 1 (Section IV) decomposes the Cloud–Edge–IoT continuum into architectural layers and cross-layers and identifies the specific capabilities required within each. This first stage is a prerequisite for prioritisation: it establishes a common analytical vocabulary, exposes cross-layer dependencies, and prevents priorities from being selected as isolated technologies detached from their architectural and operational context. Stage 2 (Section V) then consolidates those capabilities into priority clusters and analyses each cluster through a seven-dimension template and a phased transition matrix covering the 2026–2040 horizon.

Rather than predicting when each technology or policy milestone will occur, the methodology organises the capabilities, dependencies, and transition conditions that make progress plausible. This distinction is important because the Cloud–Edge–IoT ecosystem evolves through the interaction of technical maturity, regulatory obligations, deployment readiness, market adoption, and industrial capacity. The methodology also links technical cybersecurity capabilities to broader European strategic objectives: resilience, interoperability, trust, and sovereignty.

In this paper, Sovereignty Level denotes the extent to which critical capabilities and dependencies can be shaped, governed, assured, and operated under European influence

or control. It is assessed at the capability and dependency level and used to interpret the strategic significance of priority clusters, rather than to assign a rigid quantitative score.

A. STAGE 1: CLOUD–EDGE–IOT CORE DIMENSIONS (LAYERS) AND CAPABILITIES IDENTIFICATION

Stage 1 decomposes the continuum into architectural vertical and cross-cutting layers. These layers are treated as analytically distinct but operationally interdependent: a vulnerability or governance gap at one layer typically propagates to adjacent layers, so effective security requires coordinated action across the vertical and horizontal extent of the architecture [6], [24]. For each layer, Stage 1 further identifies the constituent capabilities: the technical and operational subdomains in which strategic investment, standardisation, and ecosystem building are required.

Stage 1 answers three questions:

- 1) Where are the major security and resilience problem domains within the continuum?
- 2) At which architectural layers do they emerge?
- 3) Which capabilities within each layer are sufficiently strategic, cross-cutting, or underdeveloped to justify dedicated roadmap attention?

Capabilities are derived from four sources: the architectural characteristics of distributed systems across HPC, cloud, edge, and device tiers; the recurring cybersecurity challenges observed at those tiers (hardware integrity, software supply chain verification, workload isolation, identity federation, data confidentiality, and perimeter control); cross-cutting strategic concerns (sovereignty, trust, interoperability, resilience, assurance, and regulatory alignment); and the need to connect technical architecture with operational constraints, governance requirements, and deployment pathways. A capability is therefore defined as a strategic ability that the European ecosystem should be able to design, deploy, govern, standardise, assure, coordinate, or scale in order to achieve trustworthy, resilient, and sovereign operation of the continuum. Capabilities may encompass technical controls, governance mechanisms, interoperability instruments, assurance functions, ecosystem coordination capacities, and deployment enablers.

The output of Stage 1 is a structured map of the problem space, organised into six Security Layers and four Transversal Security Cross-Layers, each populated with its constituent capabilities (see Section IV). Sovereignty-sensitive capabilities, where dependencies on external providers or assurance mechanisms are structurally significant, are identified so that Stage 2 can track their evolution.

B. STAGE 2: PRIORITY CLUSTERING AND TRANSITION MATRIX

Stage 2 groups the capabilities identified in Stage 1 into three Priority Clusters: (i) Sovereign Trusted Infrastructure, the hardware, cryptographic, and resilience foundations that anchor trust across the continuum; (ii) Cognitive Security

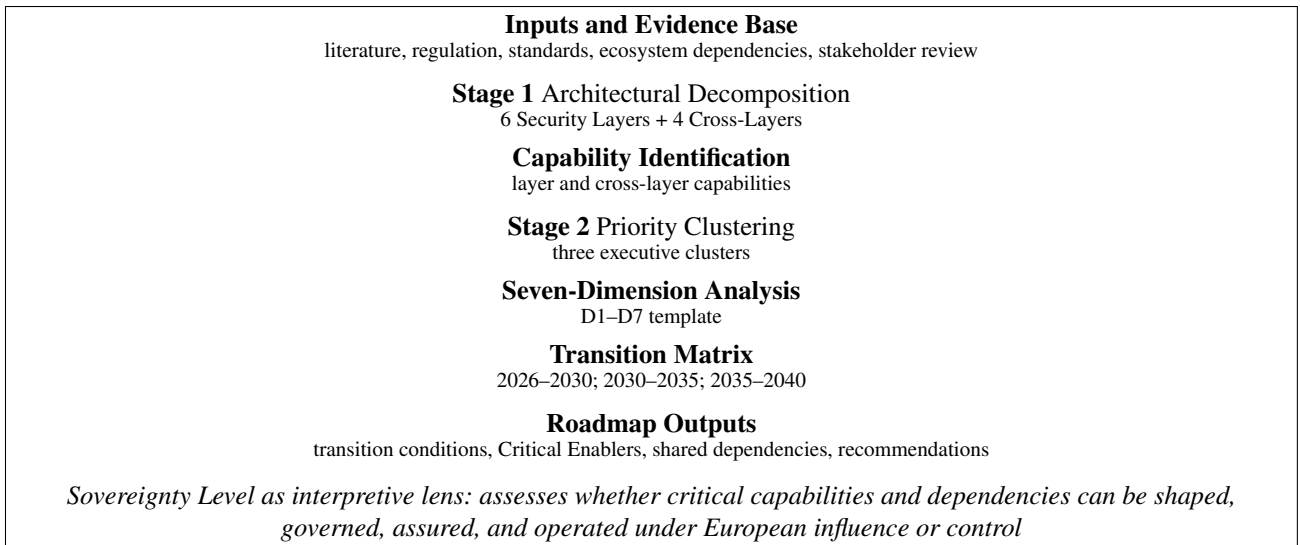


Figure 1. Two-stage methodology for constructing the cybersecurity and cyber-resilience roadmap. Stage 1 decomposes the Cloud–Edge–IoT continuum into security layers, cross-layers, and constituent capabilities. Stage 2 consolidates these capabilities into priority clusters, analyses them through a seven-dimension template, and projects them onto a three-phase transition matrix. Sovereignty Level is used as an interpretive lens across capability, dependency, and cluster assessment.

and Autonomous Cyber Defence, the AI-driven operational layer that scales defence beyond human-paced response; and (iii) Federated Operational Sovereignty, the interoperability, supply chain, and federation substrate that converts individually sovereign components into a unified European trust fabric.

The use of three priority clusters is a deliberate synthesis choice rather than an assertion that only three cybersecurity priorities exist. This is consistent with technology-roadmapping practice, where roadmaps function as strategic planning and communication instruments that organise complex technical trajectories into a smaller number of decision-relevant domains [33]. A roadmap may identify many relevant capabilities, but policy and funding coordination require higher-level priorities that communicate strategic direction, expose shared dependencies, and support portfolio-level decision-making. The three clusters therefore function as an executive structuring layer: they consolidate related capabilities into coherent investment and coordination domains while preserving the more granular capability analysis underneath.

Each priority cluster is analysed through a seven-dimension template applied consistently across the three clusters. The same template also preserves traceability to the underlying capabilities from which each cluster is derived:

- **(D1) Baseline and Target Objective**, describing the current state of the priority cluster, the intended future state, and the strategic change required to move from one to the other;
- **(D2) Primary Drivers**, identifying the threat and regulatory drivers that make the priority cluster urgent and non-deferrable;
- **(D3) High-Impact Activities**, identifying the concrete research, innovation, standardisation, or deployment ac-

tions expected to advance the priority cluster, with each action assigned to a target phase;

- **(D4) Catalysts and Enablers**, identifying the funding instruments, regulatory and standardisation mechanisms, pilot infrastructures, and ecosystem coordination mechanisms that can accelerate the priority cluster;
- **(D5) Disruptors and Systemic Risks**, identifying technical cliffs such as the emergence of cryptographically relevant quantum computers, dependency risks such as supply-chain exposure and vendor lock-in, market concentration, and geopolitical constraints that could delay, weaken, or redirect the priority cluster;
- **(D6) Failsafe Mechanisms**, specifying fallback, isolation, rollback, redundancy, and graceful-degradation mechanisms that preserve security and resilience if the primary capability or deployment pathway is delayed, compromised, or only partially realised; and
- **(D7) Transversal Meta-Layer**, assessing how the priority cluster contributes to broader European strategic objectives, including sovereignty, trust and ethics, and sustainable security.

The template provides a structured and sufficiently comprehensive overview of each priority cluster, enabling cross-cluster comparison, funding prioritisation, portfolio-level coordination, and the identification of shared dependencies.

Each cluster is then projected onto a three-phase transition matrix:

- **Phase 1: Near-Term Implementation and Pilot Actions (2026–2030)**, focused on deploying near-ready solutions, enforcing enacted regulation, validating early pilots, and closing immediate governance gaps;
- **Phase 2: Mid-Term Scale-Up, Industrial R&D and Standardisation (2030–2035)**, focused on industrialis-

ing sovereign capabilities, federating architectures, scaling validated pilots, and building the workforce and institutional infrastructure required for wider adoption; and

- **Phase 3: Long-Term Strategic Autonomy and Leadership (2035–2040)**, focused on achieving operational maturity, consolidating European leadership, advancing frontier research, and resolving long-term governance and oversight questions.

For each priority cluster and phase, the transition matrix is informed by four assessment perspectives:

- **R&I Maturity**, covering the technical maturity of the capabilities underlying the priority cluster, the maturity of their enabling dependencies, the availability of pilot deployments, and the readiness of the research and innovation ecosystem to support transition;
- **Policy, Regulation, and Standards**, covering the regulatory obligations that drive the priority cluster, the standards and certification mechanisms that support it, and the policy instruments that can accelerate or constrain adoption;
- **Operational Deployment and Adoption**, covering the availability of deployable solutions, the maturity of deployment pathways, and the readiness of users, operators, and infrastructure providers to adopt the priority cluster; and
- **Ecosystem Uptake and Strategic Impact**, covering the ecosystem conditions that shape adoption, the strategic significance of achieving the target state, and the consequences of delay or failure.

The roadmap uses transition conditions as evidence-informed progression markers indicating when movement from one phase to the next becomes plausible. In the transition matrices, these conditions are expressed through the Critical Enabler row, which identifies the technical, regulatory, standardisation, deployment, ecosystem, or sovereignty-related condition most likely to determine phase progression. A transition condition may indicate that a technical basis has matured, that a policy or standardisation basis has emerged, that a deployment pathway has been validated, that a strategic ecosystem condition has materialised, or that a Sovereignty Level improvement has become plausible through reduced dependency or increased European capacity. These conditions are not deterministic predictions or precise quantitative thresholds unless the available evidence supports such specificity.

IV. STAGE 1: CLOUD–EDGE–IOT CORE DIMENSIONS (LAYERS) AND CAPABILITIES IDENTIFICATION

The continuum is modelled as a layered reference architecture spanning four execution environments (HPC, Cloud, Edge, Devices) and organised into six *Security Layers* forming the vertical stack and four *Transversal Security Cross-Layers* serving as horizontal enablers, as illustrated in Figure 2. These environments represent distinct but intercon-

Execution environments			
HPC	Cloud	Edge	Devices
L6: Network and Perimeter			
L5: Data-Centric (PQC, TEE, QKD, SMPC)			
L4: User and Orchestration (ZT, IAM, PAM)			
L3: Application (Workloads, APIs)			
L2: Software (OS, Middleware, OTA)			
L1: Hardware (RoT, PhySec)			
CL1: Cyber-Resilient SW/AI Lifecycle			
CL2: Dynamic Risk and SecOps			
CL3: Interoperability (Simpl, CAMARA, MIMs)			
CL4: Compliance-by-Design and Assurance			

Figure 2. Security framework: four execution environments, six Security Layers (vertical stack, L1–L6), and four Transversal Security Cross-Layers (CL1–CL4).

nected sites of computation and control: HPC provides large-scale scientific and AI training capacity; Cloud offers elastic service hosting and orchestration; Edge brings computation closer to users, industrial sites, telecom infrastructure, and data sources; and Devices include resource-constrained sensors, actuators, embedded systems, and IoT endpoints operating closest to the physical world.

The decomposition aligns with authoritative reference architectures, including ETSI MEC GS 003 [22], NIST SP 800-207 and SP 800-207A for Zero Trust [4], [5], NIST CSF 2.0 [34], ISO/IEC 27400 [35], and the ENISA baseline for IoT [36]. It is also consistent with the distributed computing-continuum framing established in recent literature [1], [2] and consolidated in the European industrial technology roadmap for the next-generation cloud-edge [3].

The layers are analytically distinct but operationally interdependent: a gap at one layer can propagate to adjacent layers, as documented across the edge-computing security literature [6], [23], [24]. The model provides the shared vocabulary used throughout the paper and makes cross-layer dependencies explicit, allowing the priority clusters and transition matrices in Section V to be analysed as an interconnected portfolio rather than as isolated priorities. The EU cybersecurity certification framework [13] and its EUCC operationalisation [14] provide the regulatory anchoring for several of the capabilities defined below.

The remainder of this section introduces each layer and cross-layer by first explaining its purpose within the security framework and then listing the capabilities associated with it. Table 1 consolidates these capabilities, their layer or cross-layer codes, and their principal references. Each capability is treated as an atomic analytical unit: the ten strategic priorities analysed in Section V are derived by aggregating and translating these capabilities into actionable strategic needs.

A. SECURITY LAYERS (VERTICAL STACK)

The six Security Layers define the vertical stack of the continuum, moving from physical and silicon-level trust foundations to software, applications, identity and orchestration, data protection, and network infrastructure. They are presented from the lowest trust anchors to the communication

Table 1. Layer and cross-layer capabilities with their principal references.

Code	Capability	Primary references
L1.1	Hardware Roots of Trust; EU open-hardware (RISC-V) to EUCC “High”	[25], [26], [37]–[39]
L1.2	Physical-layer threat detection and RF fingerprinting	[6], [24]
L2.1	OS and middleware hardening; sovereign edge Linux	[4], [40]
L2.2	Secure OTA patch and update management	[41], [42]
L3.1	Workload security (service mesh, mTLS, sidecar)	[43], [44]
L3.2	In-band network telemetry and deep observability	[43], [44]
L3.3	API security gateway and policy enforcement	[40], [45]
L4.1	Zero Trust orchestration (PEP/PDP)	[4], [5]
L4.2	Federated IAM and SSI	[46]–[48]
L4.3	Privileged Access Management and JIT	[4], [5]
L5.1	Confidential computing and TEEs	[49]
L5.2	Post-Quantum Cryptography for edge/IoT	[8], [50]–[53]
L5.3	Signalling/data-plane cryptographic authentication	[54]–[57]
L5.4	Distributed reputation and trust networking (DLT)	[13], [58]–[60]
L5.5	QKD for critical links	[61], [62]
L5.6	Secure Multi-Party Computation	[12], [28], [49]
L6.1	Telco-cloud and O-RAN security	[54]–[57]
L6.2	Physical resilience and IT/OT safety	[6], [24], [54], [57]
L6.3	Edge-native autonomous guardians	[55]–[57]
L6.4	Non-terrestrial network and satellite security	[63]
CL1.1	Continuous DevSecOps	[40]
CL1.2	Model integrity and DataBOMs	[64], [65]
CL1.3	Dynamic supply-chain verification; Runtime SBOMs	[66]
CL1.4	Automated adversarial testing and purple teaming	[67]
CL2.1	Automated SOC and collaborative CTI	[68]
CL2.2	High-precision telemetry and security signalling	[69], [70]
CL2.3	Internet of Behaviour and evidence sharing	[70]–[73]
CL2.4	AI-native analytics (NWDAF, LLMSecOps)	[64], [65], [69]
CL2.5	CTI lifecycle management	[70]–[73]
CL2.6	Deception technology and active defence	[70], [71]
CL3.1	Simpl integration and heterogeneous interoperability	[58]–[60]
CL3.2	Neutral host and multi-tenant isolation	[4], [5]
CL3.3	Pan-European NaaS federation	[74]
CL3.4	Semantic security interoperability	[59], [60]
CL3.5	Cyber range and digital twin federation	[3], [13], [68]
CL4.1	PQC readiness and Zero Trust architectures	[4], [5], [50]–[52]
CL4.2	Continuous automated certification (EUCC; EUCS)	[13], [14], [75], [76]
CL4.3	Multicloud Key Management as a Service	[4], [5], [50]–[52]
CL4.4	Privacy-preserving federated computation	[28], [49]
CL4.6	Cyber risk quantification and board-level reporting	[9], [11], [34]
CL4.7	AI trustworthiness and algorithmic accountability	[12], [64], [65]

fabric because weaknesses at lower layers can undermine the guarantees provided by higher layers, while higher-layer controls are needed to operationalise security, governance, and resilience across distributed environments.

1) Layer 1: Hardware-Centric Security

Layer 1 addresses the physical and silicon-level trust foundations of the continuum, providing the verifiable anchors on which higher-layer security assurances depend. It covers hardware roots of trust, secure boot, remote attestation, Trusted Platform Modules, Physical Unclonable Functions, and physical-layer detection primitives [25], [26]. Without a verifiable silicon anchor, software-level attestation and cryptographic identity rest on assumptions that may be difficult to validate in distributed and physically exposed environments. Layer 1 is the principal locus of hardware sovereignty and the layer in which EUCC “High” certification [14] operates as a regulatory instrument of trust.

Capability L1.1, Hardware Roots of Trust. Cryptographically verifiable device and supply-chain integrity from fabrication to deployment, with a strategic focus on advancing European open-hardware designs (RISC-V) toward EUCC “High” assurance. Technical foundations derive from TPM 2.0 [25] and integrity measurement architectures [26]. European instances include OpenTitan [37], Keystone [38], and the DARE SGA1 RISC-V initiative [39].

Capability L1.2, Physical-Layer Threat Detection and Fingerprinting. Exploitation of intrinsic wireless channel characteristics, such as Channel State Information, Carrier Frequency Offset, and RSSI, to support real-time device authentication and anomaly detection at the lowest communication layer. This capability complements upper-layer cryptographic protections and can provide additional identity assurance for resource-constrained devices that cannot reliably execute conventional authentication protocols.

2) Layer 2: Software-Centric Security

Layer 2 secures the software substrate on which continuum workloads execute, including hypervisors, container runtimes, distributed operating systems, and the pipelines that maintain them. Its central concerns are attack-surface reduction through OS and middleware hardening, and cryptographically verified, failure-resilient software distribution to field-deployed nodes where physical access for remediation is limited or infeasible [41], [42]. Assurance at Layer 2 is necessary for trustworthy execution of the services and workloads built above it.

Capability L2.1, OS and Middleware Hardening. Security hardening of hypervisors (KVM/QEMU), container runtimes, and distributed operating systems, with emphasis on sovereign, lightweight Linux distributions optimised for edge resource constraints and reduced attack surface.

Capability L2.2, Secure OTA Patch and Update Management. Cryptographically verified, failure-resilient over-the-air update pipelines for field-deployed edge nodes such as smart grids, O-RAN radios, and industrial gateways, where physical access for remediation is limited or infeasible. The IETF SUI architecture [41] and the Uptane standard [42] provide the main reference frameworks.

3) Layer 3: Application-Centric Security

Layer 3 protects the workloads and service interfaces executing across the continuum, including containerised applications, serverless functions, service meshes, and the APIs that expose them. Application-layer defences are a key enforcement point for obligations that surface as runtime behaviours rather than infrastructure properties, including AI Act [12] and GDPR Article 25 [28] requirements. The NIST SP 800-204 microservices security series [40], [43], [44] and the OWASP API Security Top 10 [45] provide the main reference controls.

Capability L3.1, Workload Security. Securing containerised applications and serverless functions through hardened service meshes and sidecar proxies that provide mutual TLS, traffic policy enforcement, and granular inspection while preserving operational performance, aligned with NIST SP 800-204 [43] and SP 800-204A [44].

Capability L3.2, In-Band Network Telemetry and Deep Observability. Embedding telemetry directly into packets or service-mesh flows traversing containerised workloads, enabling real-time security monitoring and anomaly detection without relying exclusively on dedicated out-of-band monitoring infrastructure.

Capability L3.3, API Security Gateway and Policy Enforcement. A unified enforcement point for authentication tokens, rate limiting, schema validation, and payload inspection, increasingly important as containerised and AI-enabled workloads communicate through REST and gRPC APIs. This capability follows the OWASP API Security Top 10 taxonomy [45] and the DevSecOps-for-service-mesh guidance in NIST SP 800-204C [40].

4) Layer 4: User and Orchestration-Centric Security

Layer 4 governs identity, access, and policy enforcement across the multi-provider continuum. It implements continuous, context-adaptive authorisation aligned with NIST SP 800-207 [4] and its cloud-native extension SP 800-207A [5]; federated and self-sovereign identity for humans, machines, and autonomous workloads based on the W3C Verifiable Credentials Data Model [46] and, in Europe, the eIDAS 2 framework [47] and its implementing acts for the EU Digital Identity Wallet [48]; and privileged-access controls that reduce reliance on standing credentials. Layer 4 is where identity-centric and policy-driven security replace assumptions of stable network perimeter control.

Capability L4.1, Zero-Trust Orchestration. Policy Enforcement Point and Policy Decision Point architectures that enable continuous authorisation and dynamic access controls adapted to request context, device posture, workload identity, and operational risk, aligned with NIST SP 800-207 [4] and its cloud-native extension [5].

Capability L4.2, Federated Identity and Access Management. Federated and self-sovereign identity mechanisms enabling cryptographically verifiable, decentralised authentication of machines, users, and automated workloads across heterogeneous cloud-edge domains, based on the W3C Verifiable Credentials Data Model [46] and, in Europe, the eIDAS 2 framework [47] and the EU Digital Identity Wallet implementing acts [48].

Capability L4.3, Privileged Access Management and Just-In-Time Access. Reduction of over-provisioned standing privileges, a recurring enabler of advanced persistent threat campaigns, by replacing permanent access with time-bounded, contextually justified, and auditable grants.

5) Layer 5: Data-Centric Security

Layer 5 protects data across the continuum lifecycle: at rest, in transit, and in use. It spans confidential computing through Trusted Execution Environments [49]; post-quantum cryptography based on NIST FIPS 203/204/205 [50]–[52]; signalling and data-plane protection; distributed reputation and trust; Quantum Key Distribution for selected high-value links [61], [62]; and Secure Multi-Party Computation for collaborative processing across mutually distrusting parties. Layer 5 is where GDPR Article 32 [28] and AI Act Article 15 [12] obligations on confidentiality, integrity, and robustness are most directly operationalised.

Capability L5.1, Confidential Computing. Protection of data in use through Trusted Execution Environments spanning the cloud-to-IoT data path, as defined by the Confidential Computing Consortium [49].

Capability L5.2, Post-Quantum Cryptography. Migration to quantum-resistant algorithms, namely ML-KEM [50], ML-DSA [51], and SLH-DSA [52], including adaptations for resource-constrained devices exposed to long-term “Harvest Now, Decrypt Later” threats. This capability is complemented by ETSI TS 103 744 hybrid key-exchange profiles [53] and aligned with the NIS CG PQC roadmap [8].

Capability L5.3, Signalling and Data-Plane Security. Embedding trust into signalling, control-plane, and data-plane interactions, moving beyond best-effort packet forwarding toward cryptographically authenticated and integrity-protected exchanges.

Capability L5.4, Distributed Reputation and Trust Networking. Distributed reputation and trust mechanisms, potentially supported by Distributed Ledger Technology and smart contracts, enabling decentralised trust evaluation across heterogeneous providers without relying on a single governance point.

Capability L5.5, Quantum Key Distribution for Critical Links. Information-theoretically secure key exchange over optical fibre for selected point-to-point high-value links, such as government communications and critical infrastructure backbones, complementing algorithmic PQC for link-layer protection and governed by ETSI GS QKD 004 [62] and GS QKD 014 [61].

Capability L5.6, Secure Multi-Party Computation. Mechanisms that allow multiple mutually distrusting parties to jointly compute functions over combined private inputs without revealing those inputs, extending confidential computing (Capability L5.1) beyond single-party workloads to collaborative analytics across providers and jurisdictions.

6) Layer 6: Network Infrastructure and Perimeter-Centric Security

Layer 6 secures the communication fabric of the continuum, including 5G and 6G cores, disaggregated RAN, IT/OT gateways, far-edge compute, and non-terrestrial network segments. It aligns with 3GPP TS 33.501 [54], O-RAN WG11 security specifications [55], Zero Trust for O-RAN [56], the SNS JU SRIA [57], and 3GPP NTN security [63]. Layer 6 is where cybersecurity intersects most directly with physical resilience, because failures or compromises in the communication fabric can cascade into operational disruption, safety hazards, or kinetic effects in critical infrastructures.

Capability L6.1, Telco-Cloud and O-RAN Security. Hardening of distributed 5G and 6G cores and RAN components in alignment with 3GPP TS 33.501 [54], O-RAN WG11 security specifications [55], Zero Trust for O-RAN [56], and the SNS JU SRIA [57], with emphasis on secure network slice isolation, virtualised RAN protection, and sovereign operational control over critical telecom infrastructure.

Capability L6.2, Physical Resilience and Safety. IT/OT gateway security for high-availability edge environments such as smart grids, transport systems, and manufacturing sites, reducing the risk that digital breaches cascade into operational disruption, kinetic damage, or safety hazards.

Capability L6.3, Edge-Native Autonomous Guardians. Lightweight, localised security control functions, including Security-as-a-Service aggregators, hosted directly at far-edge gateways or within Open-RAN Radio Intelligent Controllers to enable autonomous enforcement at the network periphery.

Capability L6.4, Non-Terrestrial Network and Satellite Security. Security requirements for integrating satellite and non-terrestrial network segments into 6G architectures for coverage continuity, backhaul resilience, and emergency operations, aligned with 3GPP NTN specifications [63] and the EU Space Programme.

B. TRANSVERSAL SECURITY CROSS-LAYERS

The four Transversal Security Cross-Layers define horizontal security capabilities that span the vertical stack and address properties that no single layer can provide on its own: life-cycle integrity of software and AI components, operational response capacity, interoperability across providers and jurisdictions, and continuous regulatory assurance.

1) Cross-Layer 1: Automated Security Processes and Cyber-Resilient AI Lifecycle

Cross-Layer 1 spans the lifecycle of software and AI component security, from secure development and integration to runtime supply-chain verification and continuous adversarial validation. It aligns with service-mesh DevSecOps patterns [40], the NIST AI Risk Management Framework and its Generative AI Profile [64], [65], the SLSA provenance framework [66], and the adversarial robustness literature [67]. Its role is to embed cyber-resilience into how software and AI components are designed, built, shipped, verified, and tested, rather than treating security assurance as a post-deployment activity.

Capability CL1.1, Continuous DevSecOps. AI-assisted code analysis, vulnerability detection, and remediation integrated into the earliest stages of the software lifecycle, following NIST SP 800-204C [40].

Capability CL1.2, Model Integrity and DataBOMs. Protection of AI training and deployment pipelines against data poisoning, unauthorised model manipulation, and insufficient traceability of training data and model artefacts, aligned with the NIST AI Risk Management Framework [64] and its Generative AI Profile [65].

Capability CL1.3, Dynamic Supply-Chain Verification and Runtime SBOMs. Dynamic Software Bills of Materials and cryptographic proof-of-execution mechanisms that continuously verify the integrity of containerised microservices and AI components as they migrate across the multi-provider continuum, supported by the SLSA provenance framework [66].

Capability CL1.4, Automated Adversarial Testing and Continuous Purple Teaming. Continuous automated penetration testing, coverage-guided and protocol-level fuzzing, and purple-team automation against live or staging environments, providing the adversarial validation counterpart to DevSecOps and grounded in the adversarial robustness literature [67].

2) Cross-Layer 2: Dynamic Risk Management and Operational Cyber Resilience

Cross-Layer 2 governs threat detection, analytics, cyber threat intelligence, and operational response across the continuum. It supports SOC federation across Member States [68], integrates AI-native telemetry analytics via the 5G NWDAF architecture [69] and LLM-assisted security operations, and structures the cyber threat intelligence lifecycle using MITRE ATT&CK [71] and STIX/TAXII 2.1 [72], [73] as the structured-CTI foundation, complemented by ENISA threat landscape reporting [70]. Its role is to reduce detection-to-response timelines and improve coordinated operational resilience in environments where human-paced analysis alone is insufficient.

Capability CL2.1, Automated SOC and Collaborative Threat Intelligence. Federated SOC and collaborative threat-intelligence mechanisms across Member States, supported by legally protected sharing platforms that reduce institutional trust barriers and enable coordinated situational awareness [68].

Capability CL2.2, High-Precision Telemetry and Security Signalling. High-fidelity telemetry and security signalling mechanisms that feed real-time threat intelligence and detection pipelines while managing bandwidth, latency, and alert-volume constraints.

Capability CL2.3, Internet of Behaviour and Evidence Sharing. Collection, correlation, and sharing of behavioural evidence across infrastructure components to detect anomalous patterns that may be invisible to purely signature-based defences.

Capability CL2.4, AI-Native Analytics Integration (NWDAF and LLMSecOps). Integration of cybersecurity telemetry with 5G NWDAF analytics [69] and LLM-assisted security operations to support triage, correlation, investigation, and response.

Capability CL2.5, Cyber Threat Intelligence Lifecycle Management. Governance of the CTI lifecycle as a prerequisite for federated SOC operation, using MITRE ATT&CK [71] and STIX/TAXII 2.1 [72], [73], complemented by ENISA threat landscape reporting [70].

Capability CL2.6, Deception Technology and Active Defence. Deployment of decoy assets, such as honeypots, honeynets, credential canaries, and fake API endpoints, to generate high-fidelity alerts based on adversary interaction and to support earlier detection of lateral movement or reconnaissance.

3) Cross-Layer 3: Interoperable Security Mechanisms and Open-Source Verifiability

Cross-Layer 3 enables secure interoperation across heterogeneous providers, infrastructures, and jurisdictions. It anchors federation in the Simpl open-source middleware for Common European Data Spaces [58], CAMARA Network APIs for pan-European compute federation [74], semantic interoperability through OASC Minimal Interoperability Mechanisms [59] and the ETSI NGSI-LD API [60], and federated

cyber ranges for shared validation infrastructure. Its role is to convert individually sovereign components into a collectively sovereign European trust fabric while avoiding new forms of vendor lock-in.

Capability CL3.1, Heterogeneous Interoperability and Simpl Integration. Alignment with the Simpl open-source smart middleware [58] to support secure federated access, policy portability, and interoperability across Common European Data Spaces.

Capability CL3.2, Neutral Host and Multi-Tenant Isolation. Enforcement of Zero Trust edge operations across network partitions, tenants, and shared-infrastructure models, ensuring that infrastructure sharing does not imply implicit trust between operators or workloads.

Capability CL3.3, Pan-European NaaS Federation and Cross-Border Edge Roaming. Standardised interfaces leveraging CAMARA Network APIs [74] to support a federated Network-as-a-Service model for cross-border edge and connectivity services.

Capability CL3.4, Semantic Security Interoperability. Common ontologies and data models for security events, asset taxonomies, policy attributes, and risk classifications, operationalised through OASC Minimal Interoperability Mechanisms [59] and the ETSI NGSI-LD API [60].

Capability CL3.5, Cyber Range and Digital Twin Federation. Federation of national cyber ranges and digital-twin environments into shared EU-wide validation infrastructure for testing cross-provider security, resilience, and interoperability assumptions, aligned with the Cyber Solidarity Act.

4) Cross-Layer 4: Compliance-by-Design, Agile Governance and Automated Assurance

Cross-Layer 4 embeds regulatory compliance, governance, and assurance considerations into technical architecture. It operationalises continuous assurance for cloud services (EUCS, when adopted) and edge products (EUCC) [?], [13] through machine-readable control catalogues and assessment results using NIST OSCAL [75], [76]; cryptographic key management across multi-provider operations; privacy-preserving federated computation satisfying GDPR Article 25 [28]; cyber risk quantification that translates technical posture into financial and operational risk language; and AI trustworthiness and algorithmic accountability for autonomous security decisions under AI Act Article 9 [12], aligned with the NIST AI RMF [64]. Its role is to shift assurance from point-in-time audit towards continuous, evidence-based, and telemetry-supported verification.

Capability CL4.1, Post-Quantum Readiness and Zero Trust Architectures. Continuous Zero Trust verification and post-quantum readiness embedded into cloud-edge infrastructure, aligned with NIST SP 800-207 [4] and SP 800-207A [5].

Capability CL4.2, Continuous Automated Certification. Lifecycle certification tracking for cloud services (EUCS, when adopted) and edge products (EUCC) [13], [14], operationalised through machine-readable control cat-

alogues, assessment artefacts, and evidence exchange using NIST OSCAL [75], [76].

Capability CL4.3, Multicloud Key Management as a Service. Cryptographic key, policy, and lifecycle management across dynamic multi-provider computing-continuum operations.

Capability CL4.4, Privacy-Preserving Federated Computation. Cross-jurisdictional AI training and inference over distributed private datasets using privacy-preserving computation mechanisms, supporting GDPR Article 25 [28] data-protection-by-design objectives.

Capability CL4.6, Cyber Risk Quantification and Board-Level Reporting. Translation of technical security posture into financial, operational, and strategic risk language for decision-makers.

Capability CL4.7, AI Trustworthiness and Algorithmic Accountability. Assurance that AI systems making or supporting autonomous security decisions remain robust, accountable, and auditable, in compliance with AI Act Article 9 [12] and aligned with the NIST AI RMF [64].

V. PRIORITY CLUSTERS AND TRANSITION MATRICES

Stage 2 consolidates the layer and cross-layer capabilities identified in Section IV into ten strategic priorities and groups those priorities into three coordinated clusters. Each cluster is presented below through the seven-dimension template defined in Section III (D1 Baseline and Target; D2 Primary Drivers; D3 High-Impact Activities; D4 Catalysts and Enablers; D5 Disruptors and Systemic Risks; D6 Failsafe Mechanisms; D7 Transversal Meta-Layer), with per-priority sub-entries where a cluster aggregates more than one priority, and is then projected through a three-phase transition matrix spanning 2026–2040.

A. CLUSTER 1: SOVEREIGN TRUSTED INFRASTRUCTURE

Cluster 1 unifies P1 (Hardware Roots of Trust), P2 (PQC Agility), P3 (Lightweight Cryptography and Physical-Layer Security), P8 (Confidential Computing), and P9 (Resilient-by-Design Infrastructure and Island Mode). Federated policies and autonomous defence gain operational credibility in proportion to the maturity of the execution substrate. The substrate must guarantee hardware integrity through open-source silicon roots of trust, quantum-resistant confidentiality through crypto-agile schemes aligned with FIPS 203/204/205, and autonomous resilience under network or orchestration-plane compromise. The shared critical dependency is European sovereign fabrication capacity under the EU Chips Act [77], [78].

a: D1 Baseline and Target.

P1: Infineon, NXP, and STMicroelectronics achieved EUCC “High” for proprietary secure element families in 2025 [30]; fabrication remains non-EU. No EU-origin RISC-V processor has received EUCC or Common Criteria certification. Target: $\geq 60\%$ of critical infrastructure edge nodes use

EU-designed, EUCC “High” certified secure elements by 2040. P2: Edge and IoT environments depend on classical algorithms. The NIS CG roadmap [8] mandates national roadmaps by December 2026, high-risk critical infrastructure transitioned by December 2030, full completion by December 2035. Target: quantum-native continuum with full cryptographic agility. P3: Extreme-edge devices (mMTC, body-area, backscatter) are constrained in compute, energy, and memory; conventional cryptography is impractical. Target: lightweight, adaptive security mechanisms, including zero-energy physical-layer authentication. P8: Intel TDX, AMD SEV-SNP, and ARM TrustZone dominate operational TEE deployments; EU alternatives (Keystone [38], OpenTitan [37]) are at research or prototype stage. Target: end-to-end privacy-preserving AI pipelines with progressive migration to EU-origin hardware. P9: Current infrastructures assume persistent connectivity. Target: validated ≥ 72 h Island Mode across NIS2-essential and DORA-critical deployments.

b: D2 Primary Drivers.

Threat: state-sponsored exploitation of debug interfaces and side-channel attacks on field-deployed nodes; supply chain insertion; “Harvest Now, Decrypt Later” campaigns; cascading failures from cyberattacks, extreme weather, and geopolitical tensions. Regulatory: CRA Articles 10–11 with enforcement from December 2027 [10]; NIS2 Article 21(2)(e) on cryptographic policies [9]; AI Act Articles 9 and 15 on hardware integrity for high-risk systems [12]; NIS CG PQC roadmap [8]; DORA Article 11 on business continuity [11]; SNS JU SRIA on 6G resilience [57].

c: D3 High-Impact Activities.

Phase 1: fund three to five RISC-V secure element pilots under Horizon Europe Cluster 3 targeting EUCC “High” pre-evaluation; submit OpenTitan and Keystone prototypes; optimise FIPS 203/204/205 for 8/16-bit microcontrollers; deploy hybrid ML-KEM plus ECDH-P384 on the cloud side per ETSI TS 103 744 [53]; validate Island Mode soft-state protocols in 5G testbeds. Phase 2: first EUCC “High” certified EU-manufactured RISC-V secure elements in production via Chips Act pilot lines (NanoIC, FAMES); RISC-V Keystone TEEs in multi-provider Data Spaces; certified 72 h Island Mode under 3GPP/ETSI fallback profiles in cross-border 5G corridors; mandatory hybrid PQC in DORA-scoped deployments. Phase 3: EU-designed RISC-V secure elements as the hardware default; FHE end-to-end from IoT sensor to HPC compute node; zero-energy PHY authentication for battery-less mMTC; mandatory validated 72 h autonomous Island Mode for NIS2-essential and DORA-critical deployments.

d: D4 Catalysts and Enablers.

EU Chips Act pilot lines (NanoIC, FAMES, APECS, PIXEurope, WBG; combined funding $\approx \text{€}3.7\text{B}$ [77]); DARE SGA1 (EuroHPC, $\text{€}240\text{M}$) [39]; ECCC Strategic Agenda [79]; CEN/CENELEC JTC 13 hardware security standards; NIST

FIPS 203/204/205 [50]–[52] and ETSI TS 103 744 [53]; SNS JU 6G resilience projects [57].

e: D5 Disruptors and Systemic Risks.

EU Chips Act fabrication capacity shortfall: ECA Special Report 12/2025 projects $\approx 11.7\%$ global semiconductor share by 2030 against the 20% target [78]; the cancellation of Intel Magdeburg in July 2025 removed the largest EU-based advanced node commitment. Early arrival of a Cryptographically Relevant Quantum Computer before migration is complete. PQC computational overhead on battery-powered IoT (ML-KEM at $\approx 1.5\times$ ECDH on Cortex-M4 at equivalent security [80]). Side-channel breakthroughs against current TEE architectures. Insufficient cryptographic inventory of legacy OT/IoT systems.

f: D6 Failsafe Mechanisms.

TEE abstraction layers to avoid vendor lock-in during the Phase 1 dependency period; hybrid cryptographic schemes (classical plus PQC) for backward compatibility and harvest resistance; crypto-agility frameworks enabling sub-second algorithm switching; diversification across PQC algorithm families (lattice, code, hash) to avoid single-point cryptanalytic failure; autonomous node quarantine via eBPF network policy if attestation fails; layered lightweight cryptography with PHY signatures as last-resort authentication under energy starvation; redundant trust anchors, segmented autonomy zones, and tamper-resistant local decision logic for Island Mode.

g: D7 Transversal Meta-Layer.

Sovereignty: eliminates proprietary hardware dependencies and quantum decryption exposure; current TEE dependency approaches 100%. Trust and Ethics: EUCC “High” aligns with Common Criteria EAL4+, AVA_VAN.5; PQC alignment with GDPR Article 32 [28]; the 18-Member-State joint statement “Securing Tomorrow, Today” (November 2024) formalises PQC as a collective sovereignty concern. Sustainable Security: energy-efficient secure silicon for far-edge; lightweight PQC extends device lifecycles and avoids premature replacement.

B. CLUSTER 2: COGNITIVE SECURITY AND AUTONOMOUS CYBER DEFENCE

Cluster 2 encompasses P6 (LLMSecOps) and P10 (Proactive APT Management), with cross-layer dependencies on Model Integrity and DataBOMs (CL1.2), Dynamic Supply Chain Verification (CL1.3), and AI Trustworthiness (CL4.7). The cluster is structured by three inseparable sub-dimensions: AI as a security instrument (sovereign models, edge-native agents, autonomous SOC, Moving Target Defence); AI as adversary amplifier (polymorphic malware, automated exploit discovery); AI as target (training pipelines and inference endpoints as attack surfaces under AI Act Article 9).

a: D1 Baseline and Target.

P6: US-platform SIEMs dominate the EU market, no EU-headquartered vendor in the Gartner Leaders quadrant [32]. Sovereign EU security-specific foundation models are absent; OpenEuroLLM [81] has no dedicated security stream. Target: cognitive SecOps model with EU-sovereign models, auditable under AI Act Article 9, covering $\geq 50\%$ of new critical infrastructure SOC deployments by 2040. P10: Average APT dwell times in European organisations are 150–200 days. Target: ≤ 72 h dwell time in Phase 2 and ≤ 24 h in Phase 3 across NIS2-essential deployments.

b: D2 Primary Drivers.

Threat: weaponisation of generative AI for polymorphic malware, automated exploit discovery, and scaled social engineering; nation-state actors blending espionage with organised crime; memory-only execution and abuse of legitimate administrative tools. Regulatory: AI Act Articles 9, 13, and 15 creating mandatory requirements for autonomous security AI classified as high-risk [12]; NIS2 Article 21(2)(b) on incident handling [9]. Conformity assessment capacity lags obligations by approximately 24–36 months [17].

c: D3 High-Impact Activities.

Phase 1: integrate commercial LLMs into SOC workflows for alert triage and MITRE ATT&CK mapping; route NWDAF telemetry into centralised AI analytics in three to five 5G SA pilots [69]; train a first EU-sovereign SLM (≤ 10 B parameters) on GDPR-compliant security corpora using EuroHPC [82]; apply ML-based UEBA for APT IoC detection; target sub-72 h dwell time in one certified NIS2-essential pilot. Phase 2: deploy EU-sovereign SLMs at the edge for real-time SOC analytics; deception infrastructure (honeynets, credential canaries) for calibration; autonomous triage under ETSI GS ZSM 009 [83]; Zero-Trust reconfigurations per NIST SP 800-207 [4]. Phase 3: swarm-based agentic AI for decentralised threat intelligence; continuous Moving Target Defence (IP, topology, software diversity); continuous automated adversarial robustness validation.

d: D4 Catalysts and Enablers.

EuroHPC AI Factories (19 confirmed across 16 Member States; cumulative funding $> \text{€}2.7\text{B}$; IT4LIA, PIAST, MeluXina-AI list cybersecurity as a vertical) [82]; ETSI GS ZSM 009 [83]; ECCC Strategic Agenda [79]; MITRE ATT&CK [71] and STIX/TAXII 2.1 [72], [73]; ENISA threat landscape reporting [70]; NIST AI RMF and Generative AI Profile [64], [65].

e: D5 Disruptors and Systemic Risks.

Hallucinations and automation bias in safety-relevant contexts; adversarial prompt manipulation and data leakage into models; absence of a standardised adversarial robustness test suite for security AI (the single most consequential standards gap in the cluster); intelligence fragmentation and

Table 2. Cluster 1 transition matrix, Sovereign Trusted Infrastructure.

	Phase 1 (2026–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
ODP	Applied R&D to Fragmented Piloting	Standardised Integration	Ubiquitous Operation
SSL	High Foreign Dependency	Emerging EU Ecosystem	Full Sovereign Autonomy
Operational State	RISC-V secure element pilots in three to five Horizon Europe consortia; Intel/AMD/ARM TEEs dominate all operational deployments; classical cryptography on far-edge IoT; Island Mode validated in 5G testbeds by SNS JU consortia.	First EUCC “High” EU-manufactured RISC-V secure elements in production from Chips JU pilot lines; deployed in energy substations and telecom critical infrastructure in ≥ 5 Member States; RISC-V Keystone TEEs operational in multi-provider Data Spaces alongside Intel TDX; 72 h Island Mode certified in cross-border 5G corridors.	EU-designed secure elements as default in new EU ICT products; $\geq 60\%$ of critical infrastructure edge nodes on EUCC “High” EU-designed elements; FHE end-to-end across continuum; 72 h autonomous Island Mode mandatory in NIS2-essential and DORA-critical deployments.
Critical Enabler	EUCC “High” certification of an EU-origin open-source RISC-V secure element before end of phase.	Ratification of ETSI TS profiles for FIPS 203/204/205 tailored to EU deployment contexts; operational Chips JU fabrication capacity at the advanced nodes required for secure elements.	Mandatory EUCC “High” without waiver for new EU ICT products; PQC-native silicon as EU procurement default.

weak information-sharing incentives; adversarial adaptation outpacing defensive innovation; CA body capacity shortfall [17].

f: D6 Failsafe Mechanisms.

Human-in-the-loop validation for Tier-1 critical actions; bounded autonomy governed by ETSI GS ZSM 009 [83]; shadow-mode rollback if autonomous agents exhibit unstable control loops; policy-constrained orchestration with audit logging; autonomous real-time micro-segmentation to quarantine compromised components under Zero-Trust; layered detection, deception, and rapid recovery playbooks.

g: D7 Transversal Meta-Layer.

Sovereignty: reduces approximately 80% US-platform dependency in enterprise SIEM. Trust and Ethics: AI Act Article 9 compliance-by-design, explainability, and accountability for autonomous security decisions; NIS2 sector-specific implementing acts should reference MITRE ATT&CK [71]. Sustainable Security: edge-native SLMs reduce inference-time energy cost versus cloud round-trips for high-volume security telemetry.

C. CLUSTER 3: FEDERATED OPERATIONAL SOVEREIGNTY

Cluster 3 encompasses P4 (Security Interoperability via Simpl), P5 (Supply Chain Verification), and P7 (Pan-European NaaS Federation). The cluster addresses three mechanisms that transform sovereign components into a collectively sovereign continuum: federated interoperability through Minimal Interoperability Mechanisms (MIMs) in the Simpl stack [58] under Data Act portability mandates [27]; supply chain integrity through Runtime SBOMs, Digital Product Passports, and cryptographic proof-of-execution under CRA Annex I [10]; and compute federation through harmonised CAMARA Network APIs [74] and cross-border edge roaming. Supply chain verification bridges Clusters 1

and 3 because hardware attestation depends on the RISC-V secure elements matured in Cluster 1.

a: D1 Baseline and Target.

P4: European cloud-edge ecosystems remain fragmented across proprietary vendor silos with incompatible security policies. The Simpl-Open iterative MVP has been available since January 2025, with enhanced releases in June and December 2025 [58]. OASC MIM6 (Security management) remains a “Work item” without a formally published specification. Target: unified European trust fabric based on Simpl MIMs ratified as CEN/CENELEC standards and referenced in the EU Cloud Rulebook. P5: Current supply chains are opaque and globally distributed. CRA Annex I Part II mandates static SBOM disclosure from December 2027 [10]; NIS2 Article 21(2)(h) mandates supply chain risk assessment [9]; DORA Article 28 requires continuous ICT third-party monitoring [11], operationalised through the first 19 Critical ICT Third-Party Provider designations on 18 November 2025 [84]. Target: progression from static SBOM disclosure (Phase 1) to Runtime SBOMs with cryptographic proof-of-execution (Phase 2) to ZKP-verified supply chain attestation at transaction scale (Phase 3). P7: The CAMARA Fall 2025 meta-release comprises 60 APIs (10 stable) [74]; GSMA Open Gateway reaches 86 operator groups [85]. Cross-border edge compute roaming does not exist at operational scale. Target: commercial EU NaaS federation as a unified jurisdiction for data workloads, with non-EU hyperscaler market share in EU regulated cloud procurement below 40% for new deployments by 2040.

b: D2 Primary Drivers.

Threat: SolarWinds-class software supply chain attacks; disaggregation of telco clouds introducing multi-vendor complexity; the sovereignty cliff where absence of a Digital Single Market prevents EU operators from scaling against US and China hyperscalers. Regulatory: Data Act (applicable from 12 September 2025) [27]; NIS2 coordinated incident re-

Table 3. Cluster 2 transition matrix, Cognitive Security and Autonomous Cyber Defence.

	Phase 1 (2026–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
ODP	Applied R&D, Siloed AI Copilots	Standardised Integration	Ubiquitous Operation
SSL	High Reliance on US Foundation Models	Sovereign EU Edge LLMs Emerging	Global Leadership in Trustworthy AI Security
Operational State	Commercial LLMs integrated into SOC workflows for triage; NWDAF pilots in three to five 5G SA deployments; first EU-sovereign SLM under training on EuroHPC AI Factory compute; sub-72 h dwell time achieved in at least one certified pilot; no operational autonomous containment agent.	EU-sovereign SLMs deployed at the edge for real-time SOC analytics without cloud inference; autonomous triage governed by ETSI GS ZSM 009 in production; sub-72 h dwell time in certified critical infrastructure deployments; deception infrastructure operational for calibration.	Swarm-based agentic AI defending the continuum via self-evolving decentralised CTI; Moving Target Defence continuous in certified environments; ≤ 24 h dwell time across NIS2-essential deployments; EU-sovereign AI security tools in $\geq 50\%$ of new critical infrastructure SOC deployments.
Critical Enabler	CEN/CENELEC or ETSI TC CYBER publication of a standardised adversarial robustness test suite for security AI before end of phase.	Operationalisation of an AI Act Article 9 conformity assessment framework specific to autonomous security agents; activation of EuroHPC AI Factory compute contracts for EU-sovereign security model training.	International adoption of EU AI Act frameworks for trustworthy autonomous security agents by at least one non-EU partner jurisdiction.

sponse [9]; CRA SBOM obligations (December 2027) [10]; DORA CTPP monitoring [11]; the Digital Networks Act proposal (COM(2026) 16, January 2026) signals EC intent for network federation but does not mandate API standards.

c: D3 High-Impact Activities.

Phase 1: Simpl security MIM pilots in three Common European Data Spaces (Manufacturing, Health [86], Energy); deploy harmonised CAMARA APIs in two to three bilateral 5G corridor agreements; submit CRA-mandated static SBOM declarations; initiate Digital Product Passport pilots; advance ZKP verification towards demonstrator readiness. Phase 2: automated policy translation via standardised Simpl MIMs across active Data Spaces; Runtime SBOMs with cryptographic proof-of-execution in critical infrastructure in ≥ 5 Member States; automatic quarantine of unverified components in isolated network slices; CAMARA cross-border roaming scaled to ≥ 5 operator pairs; “Data Visa” mechanisms for dynamic AI model migration. Phase 3: Simpl-based security as universal Data Space standard; ZKP-verified supply chain attestation at real-time scale; operational EU NaaS federation; “Schengen for Cloud” legal framework resolving US CLOUD Act jurisdictional conflict.

d: D4 Catalysts and Enablers.

Digital Europe Programme funding for Simpl [58]; CEN-CLC/JTC 25 for data-space standards; Eclipse Dataspace Protocol; IPCEI-CIS ICRA Reference Architecture v2.0 (January 2026, total €2.6B, 19 participants, 12 Member States) [87]; EU Ecodesign Regulation Digital Product Passport; DORA Delegated Regulation (EU) 2024/1502 for CTPP assessment [11]; SNS JU [57]; Aduna as primary aggregator.

e: D5 Disruptors and Systemic Risks.

Vendor resistance to open standards; weakest-link failures in federated systems; uneven Member State compliance maturity; US CLOUD Act jurisdictional conflict [29]; con-

centration of manufacturing dependencies; limited multi-tier auditability; SME exclusion from compliance pathways; hyperscaler platform dominance through proprietary API ecosystems (approximately 70% EU cloud share) [7].

f: D6 Failsafe Mechanisms.

Autonomous unfederation or isolation of nodes failing the shared baseline; Zero-Trust architectures where federation does not imply implicit trust [4]; cryptographic attestation of policy enforcement; graceful isolation of unverified components into quarantined network slices; multi-vendor diversification; trusted fallback components; graceful degradation redirecting workloads to national infrastructure; regional autonomy zones; multi-operator fallback routing.

g: D7 Transversal Meta-Layer.

Sovereignty: breaks hyperscaler lock-in and resolves structural CLOUD Act conflict. Trust and Ethics: transparent and verifiable federation mechanisms enabling cross-border trust; the 2025 DINUM Bilan reports 70% European-provider share in French public cloud procurement market-wide and 99% within the State perimeter [88]. Sustainable Security: shared infrastructure reduces duplication of compute and certification overhead across Member States.

D. CROSS-CUTTING ENABLERS

Four dependencies span at least two clusters and require dedicated governance to avoid becoming systemic bottlenecks.

EU Chips Act sovereign fabrication (Clusters 1 and 3). Five Chips JU pilot lines are operational, with combined EU and national funding of €3.7B [77]. No pilot line is dedicated to secure element fabrication; restriction of access to any major non-EU foundry would immediately impair EU critical infrastructure attestation capability. Mitigation: strategic stockpiling, second-source agreements, and dedicated pilot lines for security-critical components.

AI Act conformity assessment infrastructure (Cluster 2). A 24–36 month gap between AI Act obligations and CA body

Table 4. Cluster 3 transition matrix, Federated Operational Sovereignty.

	Phase 1 (2026–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
ODP	Fragmented Piloting	Standardised Integration	Ubiquitous Operation
SSL	Mixed: Emerging EU Ecosystem / Hyper-scaler Dominance	Emerging EU Ecosystem	Full Federated Sovereign Autonomy
Operational State	Simpl security MIM pilots in three Common European Data Spaces with manual policy coordination; CAMARA APIs in two to three bilateral 5G corridors; no operational cross-border edge compute roaming; CRA-mandated static SBOMs submitted as point-in-time artefacts; DPP pilots initiated; hyperscalers retain approximately 70% of EU cloud market.	Automated policy translation via standardised Simpl MIMs across active Data Spaces; Runtime SBOMs with cryptographic proof-of-execution in critical infrastructure in ≥ 5 Member States; ZKP batch verification operational; CAMARA roaming across ≥ 5 operator pairs; “Data Visa” mechanisms operational; non-hyperscaler share $\geq 30\%$ in new regulated-sector procurement.	Simpl-based security as universal cross-provider Data Space standard; ZKP-verified supply chain attestation at real-time scale; operational EU NaaS federation; “Schengen for Cloud” framework enacted; non-EU hyperscaler share below 40% in EU regulated cloud procurement for new deployments; EU supply chain standards adopted in ISO/IEC and by at least two partner jurisdictions.
Critical Enabler	Simpl-Open iterative MVP consolidation with OASC MIM6 (Security management) progressing from Work item to formally published specification.	Ratification of Simpl security MIMs as CEN/CENELEC standards referenced as mandatory in the EU Cloud Rulebook; completion of DORA continuous ICT third-party monitoring operationalisation.	Enactment of the “Schengen for Cloud” legal framework resolving US CLOUD Act jurisdictional conflict.

operational readiness is documented [17]. As of March 2026, 8 of 27 Member States have designated single points of contact. Mitigation: accelerated CEN/CENELEC adversarial robustness test suite development and AI Act sandbox provisions for security use cases.

Runtime SBOM attestation (Clusters 1 and 3). Static SBOMs (CRA-mandated) are insufficient for dynamic continuum operations. Progression to cryptographic proof-of-execution and ZKP batch verification is a shared Phase 2–3 transition condition.

SME access and inclusion (all clusters). Cost barriers exclude SMEs from sovereign hardware alternatives, AI-powered SOC tooling, and Simpl MIM compliance. Digital Europe Programme co-funding should target shared-infrastructure pathways and subsidised certification programmes.

VI. STRATEGIC RECOMMENDATIONS

The 2040 destination is a secure, sovereign, interoperable, and resilient European computing continuum. Six actions follow from the analysis.

Sovereign secure silicon. Combine Horizon Europe support with the EU Chips Act to advance open-source RISC-V secure elements and pilot fabrication lines reaching EUCC “High” assurance. The 2025 certifications achieved by Infineon, NXP, and STMicroelectronics [30] demonstrate framework readiness; the remaining gap is open-source, EU-controlled fabrication.

Operationalised PQC migration. Deliver crypto-agility middleware for edge and IoT aligned with the NIS CG PQC roadmap milestones [8] and ETSI TS 103 744 hybrid key exchange profiles [53]. Make hybrid cryptographic deployment the default for new critical infrastructure.

Security interoperability through Simpl. Move OASC MIM6 (Security management) from Work item status to a published specification, then to a ratified CEN/CENELEC

standard referenced in the EU Cloud Rulebook, so that interoperability carries procurement-level enforcement.

Continuous certification. Use EUCC [14] as the foundation and unblock EUCS [13], plausibly through the 2026 Cybersecurity Act revision (COM(2026) 13) [16], to extend continuous assurance to cloud services.

Federated SOC and CTI. Build on DORA CTPP oversight [84] and cross-border CSIRT cooperation [68] to support pan-European situational awareness.

SME inclusion mandate. Implement tiered certification, shared cyber range access, and reusable compliance tooling so that NIS2, CRA, and AI Act compliance does not become a structural barrier to innovation.

VII. DISCUSSION

A. VALIDATION

The roadmap is treated as an evolving strategic artefact benefiting from expert review, stakeholder consultation, and iterative revision. Validation is conducted through expert workshops within the NexusForum and EuCloudEdgeIoT Working Group on Cybersecurity; consultation with national cybersecurity agencies and industry; and policy dialogue with European Commission services. Validation addresses completeness and relevance of the architectural decomposition, clarity and non-overlap of capability and priority formulations, coherence of priority clustering, credibility of transition conditions, and reasonableness of Sovereignty Level assessments.

B. DISCUSSIONS, LIMITATIONS AND FUTURE WORK

The methodology is intended to support strategic agenda setting, expert discussion, cross-priority comparison, and iterative roadmap refinement under conditions of ecosystem uncertainty. Its value lies in providing a shared analytical structure through which stakeholders can align research, policy, deployment, and investment decisions around a common set of objectives.

At the same time, the roadmap should not be interpreted as a precise forecast of future cybersecurity developments. The Cloud–Edge–IoT landscape is shaped by rapidly evolving technologies, regulatory implementation, market dynamics, and geopolitical conditions, all of which may alter the trajectory of the priority clusters. The analysis therefore offers a strategic synthesis of the state of the art and expert judgement, rather than a deterministic prediction of how the cybersecurity landscape will evolve.

Future work could extend the roadmap in three directions. First, the architectural analysis could be instantiated in sector-specific use cases, such as energy, healthcare, finance, and public procurement, to examine how cybersecurity priorities are affected by sector-specific regulation, procurement rules, supply-chain constraints, and market structure. This would help capture dependencies that fall outside a cybersecurity-centred perspective. Second, the Sovereignty Level construct should be further operationalised by identifying measurable proxies for strategic dependency, such as provider concentration, certification availability, fabrication capacity, and cross-border deployability. Third, the roadmap should be maintained as an iterative and dynamic instrument, periodically revised as new empirical evidence, stakeholder feedback, regulatory milestones, and technological developments become available.

VIII. CONCLUSION

The Cloud–Edge–IoT continuum requires Europe to move beyond reactive and fragmented cybersecurity models towards autonomous, federated, and certified Security-by-Design. The structured methodology presented here identifies security dimensions, derives strategic capabilities, and consolidates them into priority clusters aligned with resilience and digital sovereignty objectives. The ten priorities form an interdependent strategic portfolio that must be developed in parallel. The 2025–2026 period has produced concrete milestones (EUCC “High” certifications across all three major EU secure element manufacturers, a binding PQC migration framework, DORA operationalisation through CTPP designation, and emerging Simpl infrastructure) and persistent gaps (NIS2 transposition delays, EUCS deadlock, contested semiconductor targets, continued hyperscaler dominance at approximately 70% of the EU cloud market). The ambition–implementation gap defines where the coordination challenge is most acute and where strategic investment will yield the greatest returns. Future work should extend this analysis through empirical validation of Sovereignty Level assessments, sector-specific instantiation (energy, healthcare, financial services), and integration with the next EU Framework Programme funding landscape.

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